

SEQUENCE DIAGRAM (FYP)

SCENERIO : Patient Uses AI Health Assistant for Symptom Check

Overview

It shows how different system components interact when a patient enters symptoms and receives AI powered health advice along with triage recommendations.

System Components (Participants)

- Patient: The user interacting with the system.
- Chat UI: The user interface where patients enter symptoms.
- Triage System: Evaluates urgency and applies medical rules.
- Backend API Database: Manages authentication, data storage, and API requests.
- AI Service: Analyzes symptoms using medical knowledge.
- NLP Engine: Processes natural language text and identifies medical entities.
- Knowledge Base: Stores medical information, conditions, and guidelines.

Explanation

Phase 1: Initial Input & Authentication

The patient enters symptoms into the Chat UI.

The Chat UI sends a POST request to the Backend API with:

- userId: Patient identifier
- message: Symptom description
- sessionId: Current conversation session
- language: User's preferred language

The Backend API authenticates the request using JWT Token (JSON Web Token for security).

Phase 2: Message Processing

The Backend forwards the message with session context to the Triage System.

The Triage System sends the text to the NLP Engine for processing.

Phase 3: NLP Processing

The NLP Engine processes the symptom text and:

- Extracts entities (identifies symptoms and severity)
- Parses the medical information

- Returns parsed symptoms back to the Triage System.

Phase 4: Knowledge Base Query

The Triage System queries the Knowledge Base for symptom information.

The Knowledge Base returns:

- Related medical conditions
- Clinical guidelines

Phase 5: AI Analysis

The Triage System sends symptoms and context to the AI Service.

The AI Service analyzes symptoms using the medical knowledge base.

The AI generates advice and returns it to the Triage System.

Phase 6: Triage Assessment

The Triage System applies triage rules to assess the urgency level.

It creates a response and triage level classification.

Phase 7: Data Logging

The Backend API logs the interaction to the database for record keeping.

Phase 8: Response Delivery

The Backend returns a response to the Chat UI with:

- Message: AI generated advice
- TriageLevel: Urgency classification
- Recommendations: Suggested next steps.

Phase 9: Display Results

The Chat UI shows health advice and next steps to the patient.

Phase 10: Triage-Based Recommendations (Alternative Flows)

The system provides different recommendations based on urgency:

- High Urgency (Red/Emergency)
Recommend immediate medical attention.
- Medium Urgency (Yellow/Warning)
Suggest consultation within 24-48 hours.
- Low Urgency (Green/Safe)
Provide self-care recommendations.

Key Technical Features

Security

JWT Token Authentication: Ensures only authorized users can access the system.

Secure API endpoints.

Multi-language Support

The system accepts language parameters (English, Urdu, etc.).

NLP processes text in multiple languages.

Data Flow

Synchronous communication (solid lines): Direct request-response.

Asynchronous processing (dotted lines): Background operations.

Database Operations

All interactions are logged for:

- Medical history tracking
- System improvement
- Legal and compliance purposes.

System Architecture Benefits

Modular Design: Each component has a specific responsibility.

Scalability: Components can be scaled independently.

AI Powered: Combines NLP and AI for accurate analysis.

Safety First: Triage system ensures urgent cases are prioritized.

Knowledge Based: Uses medical databases for reliable information.

Parameters Shown in Diagram

According to the document requirements, key parameters include:

User IDs (patientId, doctorId)

Session tokens (JWT)

Timestamps (ISO 8601 format)

Status codes (200, 400, 500)

Language parameters ("urdu", "english")

Confidence scores (0.0 - 1.0)

Urgency levels (Green/Yellow/Red/Emergency).

Use Case Example

Scenario: Patient enters "I have chest pain and shortness of breath."

NLP extracts: symptoms = ["chest pain", "shortness of breath"], severity = "high."

The Knowledge Base identifies: possible conditions = ["cardiac event", "anxiety", "respiratory issue"].

AI analyzes and generates advice.

The Triage System classifies this as HIGH URGENCY.

The patient receives: "Seek immediate medical attention - call emergency services."

This diagram demonstrates a complete, production-ready AI health assistant system with proper error handling, security, and medical safety protocols.